

Entity-Centric High-Recall Retrieval for Scientific “Which Methods” Questions: A Development-Only Redesign (V2, Second Iteration)

Experiment: `entity_centric_high_recall_v2`

Working tree: `clean_tshc_v2` Primary system: A7 Label: DEVELOPMENT

2 October 2026

Abstract

This report documents a frozen, development-only redesign of an entity retrieval pipeline over a materials-informatics knowledge graph snapshot. The prior V1 system and a first V2 attempt both failed at the *candidate-admission* stage: the correct answer entities were usually absent from the top of the candidate list, so no reranking strategy could repair the outcome. The redesign therefore targets admission first: canonical entity grouping with graph-provided aliases and type families, a separately indexed evidence store derived from inferred schema roles, *six* retrieval lists (dense over the raw question, dense over a structured view, BM25, safe name matching, evidence-dense, and bounded graph expansion — five channel families if the two dense views are counted as one), a deterministic union, and reciprocal rank fusion. A genuine MS MARCO relevance cross-encoder reranks a fixed prefix, and a pinned local natural-language-inference model is used as an explicitly uncalibrated *requirement-verification proxy* with four-status semantics over a stable top-20 tiering. Every entity channel applies the same **hard** answer-family type filter; there is no soft type prior in this pipeline. The predeclared hypothesis is **partly supported**. Candidate recall rose decisively: the A5 fusion arm admits 37/49 directly mapped target occurrences by depth 100, 44/49 by depth 500, and 46/49 in the full union, against 3/49 at depth 100 for the previous V2 high-recall arms. Final ranking did not follow proportionally. The declared primary system A7 recovers 10 of 63 target occurrences at top-10 (15.87%), against 3/63 for the previous V2’s dense baseline and 2/63 for its relevance arm, but a predeclared smaller-pool control, A6_100, reaches 12/63; mean reciprocal rank is *higher* for the simple round-robin union A4 (0.2013) than for A5 (0.1554), A6 (0.1480) or A7 (0.1516). Among the 44 target occurrences admitted to the 500-deep reranking pool, 18 moved up and 26 moved down, with the conditional median rank moving from 39 to 50.5. High recall did not deliver proportional ranking quality, and the larger reranking pools were worse on top-10 recovery here. All numbers are development diagnostics from frozen artifacts of a single seeded run; no held-out, variance or inferential-superiority claim is made.

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1 Motivation

1.1 What V1 and the first V2 established

The task is a small set of expert-authored scientific questions of the form “Which methods (by Feb 2026) are best suited for ...” and “What are the leading ...”, each annotated with one or more target answer entities. The retrieval corpus is a knowledge graph snapshot containing typed nodes (`method`, `technique`, `component`, `cited_work`, `dataset`, `metric`, `article`) and typed relations.

V1 used a single dense bi-encoder over a bounded textual rendering of each graph node. Its failure mode was not subtle ranking error but outright absence: in the current matched re-run, the V1-like control A0 admits only 6 of 49 directly mapped target occurrences anywhere in a 500-candidate list, and recovers 0 at top-10. The first V2 iteration, archived under `archive/flat_20261001`, attempted richer scoring without fixing admission; its high-recall arms (V2-B, V2-C) admitted only 3 of 49 mapped occurrences at depth 100 and recovered 1 and 2 occurrences respectively at top-10, with MRR 0.0168 and 0.0566, while its own dense baseline arm recovered 3.

Two lessons drive this redesign. First, *admission constrains everything downstream*: each later component is conditioned on a pool it cannot extend, so recall at realistic depths must be measured and reported before any reranking claim. Second, *rank statistics must be explicit about misses*. The archived arms’ rank means were computed over the full corpus, where every target necessarily has some rank and the miss count is zero by construction. Those figures are valid under that protocol; they are simply a different quantity from the conditional means over a truncated candidate list reported here, and the two are not comparable without also reporting misses.

1.2 Development-only status

The benchmark questions and their gold annotations were already inspected by the author during V1 development, and that exposure motivated this redesign. Separating prediction from evaluation at execution time (Section 11) removes *execution-time* leakage: the predictor cannot read gold files, historical outputs, reports, notes, or the evaluator. It does not and cannot remove the fact that the benchmark had already influenced the design and the request. No claim is made here about the precise extent of any design-time exposure beyond that. Consequently:

- every number in this report is a **development diagnostic**;
- no claim of inferential or statistical superiority over V1, prior V2, or any external system is made or justified;
- with 14 scored questions and 63 target occurrences, differences of one or two occurrences are at the resolution of the measurement and are reported as observations, not as effects.

1.3 Predeclared hypothesis

Hypothesis. Replacing single-channel dense node retrieval with entity-centric canonical grouping, a separately indexed evidence store, multi-channel deterministic union and rank fusion will (H1) raise candidate recall of directly mapped target occurrences at operationally useful depths by a large margin, and (H2) that this improved admission, combined with a genuine relevance cross-encoder and a requirement-verification reranking stage, will translate into materially better top-10 identity recovery.

Verdict: partly supported. H1 is supported *by the observed development evidence* on candidate recall (Section 12.1); the measurement is a single run on a design-exposed question set,

so the support is for the recall claim as measured, not an unqualified general result. H2 is mixed. Top-10 recovery rose from 3/63 (previous V2 dense baseline) and 2/63 (previous V2 relevance arm) to 10/63 for A7, but that overall comparison involves a redesigned index, new channels and new fusion, and therefore does not isolate the contribution of the cross-encoder. Within the matched new run the incremental reranking picture is mixed: top-10 hits move 6 (A5) $\rightarrow$ 9 (A6) $\rightarrow$ 10 (A7) while MRR moves 0.1554 $\rightarrow$ 0.1480 $\rightarrow$ 0.1516, and a predeclared smaller-pool control beats the declared primary on top-10.

2 Data and time semantics

All current questions are evaluated against the frozen **2026 snapshot**. The index built for that snapshot contains 2978 canonical entities, 4274 evidence items, 10241 entity–evidence links and 6368 inferred role arcs, with 0 rejected edges and 2508 entities possessing *no* candidate-specific evidence (Table 1). These are **index object counts** — entities, evidence items, links, arcs — and must not be read as target-occurrence counts; the occurrence-based counts used for scoring are defined in Section 11.1.

The 2508 figure (84.2% of entities) bounds what the evidence-dependent machinery can do: for those entities requirement verification has no eligible candidate-specific premise. Article or structural context may still provide evidence-retrieval links or cross-encoder context; otherwise the card contains identity alone. They remain retrievable through the two dense channels, which score bounded renderings of their member nodes, and through BM25, whose per-entity lexical document always contains the entity’s names and graph aliases (Section 7.2).

Table 1: Frozen 2026 index statistics. These are index object counts, not scored occurrences.

Quantity	Value
Canonical entities	2978
Evidence items	4274
Entity–evidence links	10241
Inferred role arcs	6368
Rejected edges	0
Entities without candidate-specific evidence	2508

Time semantics are deliberately conservative:

- **first_seen_year** defines *publication visibility within this graph*: an entity is visible to a query with cutoff y only if **first_seen_year** $\leq y$. It is the first year the graph records the entity, which is not a guarantee of the actual date of discovery or first use.
- **last_seen_year** is **not** an expiry date. An entity last observed in 2021 is not removed from a 2026 query. Only its *undated aliases* are affected: because alias annotations carry no individual dates, aliases contributed by a member node are suppressed when that node’s last observation **exceeds** the snapshot cutoff being constructed. The rule therefore only bites when the parser derives a cutoff earlier than the node’s last observation; for a 2026 query over the 2026 snapshot nothing is suppressed by it.
- When the parser derives an earlier year cutoff, an edge must pass *both* the edge-year check and the asserting-article-year check; an edge whose own year is admissible but whose asserting article postdates the cutoff is rejected.
- Month precision cannot be recovered from year-only evidence. Questions phrased “by Feb 2026” are therefore answered with year granularity. This is an unavoidable approximation, not a modelled constraint.

3 Canonical entities, aliases and type families

Surface normalization applies, in order: Unicode NFKC, casefolding, whitespace collapse, and typographic-hyphen normalization to ASCII hyphen. All other punctuation is *preserved*. Writing $\nu(\cdot)$ for this normalizer, two member nodes m_i, m_j merge into one canonical entity only if

$$\nu(\text{name}(m_i)) = \nu(\text{name}(m_j)) \quad \wedge \quad \mathcal{F}(\text{type}(m_i)) = \mathcal{F}(\text{type}(m_j)), \quad (1)$$

where $\mathcal{F}$ maps a node type to its family. The *approach* family is **method**, **technique**, **component**, and **cited_work**; **dataset**, **metric** and **article** each form their own singleton family. Equation (1) requires *identical* normalized strings: no fuzzy, embedding-based or abbreviation-based merging is performed, because an inferred merge would assert an identity the graph does not.

The canonical identifier of an entity is derived from the **lowest member node ID**, never a gold label and never a human-chosen preferred name, which keeps the identifier space independent of the annotation used later for evaluation.

Aliases are taken from the graph only. Actual graph aliases are used as *retrieval strings* — they enter the name channel and the BM25 lexical document — but they never induce a merge under Equation (1). An alias relation in the graph is evidence that a string refers to an entity; string similarity between two entity names is not.

3.1 Hard answer-family filtering

Answer type families map a question’s extracted answer head(s) to admissible entity types: **method**, **model**, **architecture**, **algorithm**, **tool** and **system** map to the approach family; **technique** maps to (**technique**, **method**, **component**); **benchmark** maps to (**dataset**, **metric**); **dataset** and **metric** map to themselves; and **article**/**paper** map to (**article**, **cited_work**).

This filtering is **hard, and it is applied identically by every entity channel**. A single eligible set is computed once per question as

$$\mathcal{A}(q) = \{E : \text{types}(E) \cap \bigcup_{t \in \text{heads}(q)} \mathcal{Fam}(t) \neq \emptyset\}, \quad (2)$$

with the convention that an empty or unmapped head set admits everything. The two dense channels, BM25, the name channel, the evidence channel and the graph expansion all restrict their output to $\mathcal{A}(q)$. **There is no soft type prior anywhere in this pipeline.** The configuration file retains a legacy key `soft_type_prior = 0.8`; it is *unused* by this implementation and is reported here only so that a reader inspecting `config.json` is not misled. Consequently **TypedRaw** is not “A1 plus hard filtering”: both arms are hard filtered, and **TypedRaw** isolates the *raw-question dense view alone* under that filter, while A1 takes the maximum over the raw and structured dense views under the same filter. The only unfiltered arm is A0, which is the V1-like dense control over graph nodes.

The hard filter has a measured cost: three **EXACT** target occurrences in Q14 (JARVIS-DFT, AFLOW, Alexandria) are excluded (Section 12.6). Q14 asks for “datasets and benchmarks”, whose families are $\{\text{dataset}, \text{metric}\}$; the graph nodes bearing those three names carry node types outside that set, so Equation (2) removes them before any channel runs. The honest description is therefore that these are scientific targets the question requests *as* datasets/benchmarks whose *graph typing* differs from the requested families; no claim is made here about what the graph does type them as, because that was not inspected for this report.

4 Query representation

The **original question is retained verbatim**, including every modifier, and it is the string used as the relevance query for the cross-encoder. Extractive structure is strictly supplementary. A deterministic extractive parser produces:

- the answer head(s) and the corresponding type families;
- *preserved pre-head scientific modifiers* — e.g. “universal” in “universal MLIPs”, “twisted van der Waals” in Q3 — kept attached rather than stripped as stopword-like qualifiers, since in this domain the modifier usually carries the discriminating content;
- coordinated answer families, so that “datasets and benchmarks” in Q14 yields a union of admissible families rather than an arbitrary single choice;
- core requirements, additional requirements and exclusions, used only by the verification stage.

Two query views are scored independently: the original question q_o and a structured rendering q_s . A third, parsed-only view q_p is encoded solely to drive the **ParsedOnly** control and never enters the union. **ParsedOnly** is markedly worse than the original-question arms (MRR 0.0843 versus 0.1666 for A1). The supportable reading is narrow: on these 14 questions, with this encoder and this rendering, replacing the raw question by the structured rendering lost ranking quality, which is why the raw question is retained. It does not establish that extractive parsing destroys information in general, and indeed the structured dense view recovers slightly *more* mapped occurrences than the original-question view (21 versus 19, Table 5).

5 Indexed evidence and role inference

5.1 Separate evidence encoding

Every evidence node is encoded *separately* by type: **claim**, **task**, **problem**, **technique**, **component**, **dataset**, **metric**, **article**. Each entity links to its evidence with source IDs, relation paths, provenance and candidate-specific flags. Encoding per node rather than concatenating an entity’s evidence into one document is a deliberate design choice: a question typically matches one specific claim or task, and concatenation would average that signal into the rest of the entity’s text.

Method–object roles are inferred from **unique member type signatures**, never from member order in the underlying tuple. The supported patterns are:

1. one **method** plus typed **task/problem/technique/ component/dataset/metric** members;
2. one **article** plus one **claim**, optionally plus one **method**;
3. one **article** presenting one **method**;
4. one **article** citing **cited_work** targets.

Anything ambiguous — in particular **extends** relations with an unresolvable direction, or relations whose member types do not match a supported signature — is omitted and recorded as a rejected edge. This is **schema inference**, not annotated linguistic role labelling and not entailment. A role arc states that the graph’s typed structure is consistent with a reading, not that the reading is semantically verified.

5.2 Safe mention attachment

Claims are attached to entities by exact, bounded mentions of entity names or graph aliases. Short generic words are excluded: a string is admissible as a mention anchor only if it is at least three characters and either contains digits, is CamelCase, is all-caps of length ≥ 3 , or is a multiword name of at least ten characters. This blocks the pathological case in which a generic token used in ordinary prose attaches unrelated claims. Article titles supply provenance context for an entity; they do *not* license attaching all claims from the same article to that entity.

The evidence index itself is **uncapped** — its purpose is recall. Per-family caps apply only when selecting text for the cross-encoder card (Section 8).

6 Bounded graph expansion

Graph generation admits exactly two path shapes:

$$\text{evidence} \longrightarrow \text{method} \quad (1 \text{ hop}), \quad (3)$$

$$\text{evidence} \longrightarrow \text{article} \longrightarrow \text{presented/cited entity} \quad (2 \text{ hops}). \quad (4)$$

Only **article** nodes may be intermediate. There is no author traversal, no unrestricted spreading activation, no recursion and no hierarchy beam. Parameters are fixed: top-200 evidence seeds, article degree cap 80, fanout cap 64, maximum 2 hops, path decay $\gamma = 0.8$. A path p of edge length $\ell(p)$ from seed s_p with seed score $\sigma(s_p)$ contributes

$$g(E) = \max_{p: s_p \longrightarrow E} \gamma^{\ell(p)-1} \sigma(s_p), \quad \gamma = 0.8, \ell(p) \leq 2, \quad (5)$$

and only destinations in $\mathcal{A}(q)$ are kept, so the hard type filter applies to graph expansion as well. Each surviving entity exports up to eight scored path traces.

Two limits must be stated explicitly. First, **direction inference is limited**: where the graph does not unambiguously assert which of two entities extends or specializes the other, the arc is dropped rather than guessed, so some legitimate paths are simply unavailable. Second, **citation paths are not entailment**. That article A cites work W places W in A ’s structural context; it says nothing about whether W satisfies the question’s requirements. Accordingly, citation-derived evidence is admitted as structural context for *retrieval* but is categorically **ineligible as a premise for requirement verification** (Section 9).

7 Retrieval channels and fusion

7.1 Dense channels

The dense encoder is BAAI/bge-small-en-v1.5 at revision

5c38ec7c405ec4b44b94cc5a9bb96e735b38267a, used with **no query instruction prefix**, exactly as in V1, so that the comparison isolates architecture rather than prompting. The document representation retains the old bounded V1 node rendering for the same reason.

Each eligible canonical entity E is scored by the **maximum** over its member nodes, for each query view independently:

$$s_{\text{dense}}(E, q) = \max_{m \in E} \frac{\mathbf{v}(q) \cdot \mathbf{v}(m)}{\|\mathbf{v}(q)\| \|\mathbf{v}(m)\|}. \quad (6)$$

Max-pooling rather than mean-pooling was chosen so that a canonical entity with many weakly related member nodes is not penalized relative to a singleton. That is the design motivation; this study did not run a mean-pooling variant, so the motivation is not an empirically demonstrated mechanism here. Arm A1 combines the two views by maximum:

$$s_{A1}(E) = \max(s_{\text{dense}}(E, q_o), s_{\text{dense}}(E, q_s)). \quad (7)$$

A0 is the V1-like control: raw-question dense retrieval over all non-author graph nodes, deduplicated to an answer entity when one exists, with no canonical grouping and no type filter.

7.2 Lexical and name channels

BM25 uses $k_1 = 1.5$, $b = 0.75$. Each entity has one lexical document, formed by concatenating its names, its graph aliases, and the text of each unique indexed evidence item once.¹ For query term set Q and document d of length $|d|$ with average length $\bar{L}$:

$$\text{BM25}(q, d) = \sum_{t \in Q} \text{IDF}(t) \frac{f(t, d) (k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \frac{|d|}{\bar{L}}\right)}. \quad (8)$$

The query string is q_o concatenated with q_s . Only eligible rows with *positive* term overlap may enter the BM25 top-500; admitting zero-overlap rows would let arbitrary tie-broken entities cast rank votes in the fusion step.

Safe name matching is a *separate* channel from BM25, so that exact-identity opportunity is measurable on its own. Its measured contribution here is zero (Table 5), which is itself a finding: the questions almost never contain the target’s surface name, and 46 of 63 target occurrences carry the secondary observation `NO_NAME_MATCH`.

7.3 Evidence-dense and graph channels

The evidence channel takes the maximum cosine across the two query views for each evidence item, selects the top 500 evidence seeds, maps each seed to the eligible answer entities it supports through the inferred role arcs, and max-pools per entity:

$$s_{\text{ev}}(E) = \max_{i \in \text{Ev}(E) \cap \text{Seeds}_{500}} \max(\cos(q_o, i), \cos(q_s, i)). \quad (9)$$

The graph channel consumes the same ranked seed list but is kept as a separate list so that its contribution is separately attributable and separately ablatable.

7.4 Deterministic union and reciprocal rank fusion

Six ranked lists are available for fusion: `dense_original`, `dense_structured`, `bm25`, `name`, `evidence` and `graph`. Counting the two dense views as one family gives five channel families. Every list contributes **at most 500 entity IDs**. Arm A2 round-robins the first four lists; A3 adds evidence; A4 adds graph. Round-robin interleaving is a *union construction with a deterministic order*; it uses no score calibration at all, which is why it is a useful baseline against fusion.

A5 uses the **identical A4 union** and changes only the ordering, to reciprocal rank fusion:

¹The frozen specification phrases this as scoring names, aliases and evidence “independently”; the implementation realizes all three string sources inside a single concatenated per-entity document, which is what is described and measured here.

$$\text{RRF}(E) = \sum_{c \in \mathcal{C}} \frac{1}{k + r_c(E)}, \quad k = 60, \quad (10)$$

where $r_c(E)$ is E 's rank in list c and lists that did not retrieve E contribute nothing. Equation (10) uses *only rank contributions*; raw channel scores are never mixed, because dense cosines, BM25 scores and decayed path weights are not on a common scale. Ties are broken by canonical entity ID, making the output a deterministic function of the union. Because A4 and A5 share their union exactly, every difference between them is attributable to ordering alone — and that difference is *not* uniformly favourable (Section 12.2).

Each exported candidate carries all of its contributing channel ranks plus the selected evidence and path reasons, so that every position in the final list is auditable.

8 Query-dependent cross-encoder input

The reranker is `cross-encoder/ms-marco-MiniLM-L6-v2` at revision `233902d25c440f23af6f7d6e94d2946bac0bee0a` — a **genuine relevance** cross-encoder, trained for query–passage relevance, not an entailment model repurposed as a relevance scorer.

Card construction, within each evidence family: sort by maximum original/structured cosine with deterministic tie-breaks, deduplicate normalized text, apply the family caps (claims 4, explicit mentions 3, tasks 2, problems 2, techniques 2, components 2, datasets 1, metrics 1, articles 1, graph evidence 2), and **preserve candidate identity first**. The block order is fixed: identity, then claims, explicit mentions, tasks, problems, techniques, components, datasets, metrics, article titles, structural path evidence.

Token packing retains the identity block, then the strict *prefix* of remaining blocks that fits within 512 tokens — a prefix, not a greedy best-fit selection, so that packing is a deterministic function of the fixed block order. The query is the full original question; only an overlong query would be truncated, and identity is preserved if so. Included blocks, omitted blocks and token counts are exported for every pair.

Two properties are enforced. Relevance logits are **not probabilities** and are never interpreted as calibrated confidences. And the **frozen pool cannot change during reranking**: the reranker reorders its prefix and the untouched RRF tail is appended unchanged, so reranking can never add or remove a candidate.

9 Requirement verification

For each of the top 20 candidates, and for each core requirement, additional requirement and exclusion, the maximum-cosine item is retrieved from the **full candidate-specific evidence index**, not merely from the token-truncated cross-encoder card. Article-level and structural-path evidence are **ineligible** as premises. The pinned local model `cross-encoder/nli-deberta-v3-small` at revision `fa2804872c3b4bd748f38c0185cc85775361e735` scores the selected premise against a literal candidate-specific hypothesis, and the entailment/neutral/contradiction probabilities are recorded as **uncalibrated proxies**.

With threshold $\tau = 0.7$, let (p_e, p_c) be the entailment and contradiction probabilities for a positive component, and let them be *swapped* for an exclusion. Then

$$\text{status} = \begin{cases} \text{SUPPORTED} & \text{if } p_e \geq \tau \text{ and } p_c < 1 - \tau, \\ \text{CONTRADICTED} & \text{if } p_c \geq \tau \text{ and } p_e < 1 - \tau, \\ \text{NOT_SUPPORTED_BY_AVAILABLE_EVIDENCE} & \text{otherwise, including no eligible premise,} \\ \text{NOT_EVALUABLE} & \text{if a premise exists but the model is unavailable.} \end{cases} \quad (11)$$

The swap is load-bearing for interpretation: for an *exclusion*, high entailment of the excluded property counts *against* the candidate and high contradiction counts in its favour. A **SUPPORTED** status on an exclusion component is therefore backed by a high *contradiction* probability, so the aggregate **SUPPORTED** count in Section 12.7 cannot be read as “all of these had $P(\text{entail}) \geq 0.7$ ” unless exclusion components are known to be absent.

Two further semantics matter. **Missing evidence receives the same unsupported status as neutral evidence, never contradiction**: the absence of a premise is not a refutation. And **NOT_EVALUABLE** exists specifically so that an unavailable model yields an explicit non-result where evidence does exist, rather than a silent zero indistinguishable from a negative finding.

A7 applies a **stable three-tier reordering within the top 20**: all requested components proxy-supported first (tier 0); unresolved with no contradiction next (tier 1); any proxy contradiction last (tier 2); the original cross-encoder order breaks ties inside each tier. Every entity outside the top 20 is left unchanged. This is a heuristic reranking experiment, not scientific entailment and not calibrated truth. Unsupported evidence alone is never treated as false. The NLI model can and does misread conditions and negation, so Section 12.3 reports harms alongside gains.

10 Experimental arms

Table 2: Arms and controls. “Union” = candidate set construction; “Order” = final ordering rule. All arms except A0 apply the hard answer-family filter of Equation (2).

Arm	Order	Construction
A0	dense	V1-like: raw-question dense over all non-author nodes, no canonical grouping, no type filter.
A1	dense	Canonical entities, max over original and structured dense views.
TypedRaw	dense	Raw-question dense view alone, same hard filter; isolates the raw view.
ParsedOnly	dense	Parsed-only query view alone; isolates loss from restructuring.
A2	round-robin	Union of original-dense, structured-dense, BM25, name.
A3	round-robin	A2 + evidence channel.
A4	round-robin	A3 + graph channel.
A5	RRF ($k = 60$)	Identical union to A4 ; ordering changed only. Primary first stage.
WithoutEvidenceRRF	RRF	A5 ablation: evidence list removed from fusion.
WithoutGraphRRF	RRF	A5 ablation: graph list removed from fusion.
A6 (=A6_500)	CE prefix 500	Relevance CE over first 500 of A5; unchanged RRF tail appended.
A6_100, A6_200	CE prefix 100/200	Predeclared pool-size controls; <i>reuse identical pair scores from the 500 run</i> ; unchanged tail appended.
A7	CE + tiers	Declared primary : requirement verification over A6’s top 20 only.

The pool-size controls A6_100 and A6_200 were predeclared before results were seen. They

reuse the *same* pair scores computed in the 500 run; their recorded prefix timings (Table 15) are therefore **amortized estimates, not independently benchmarked experiments**. No diffusion arm was run.

The declared primary system remains A7. As shown below, A6_100 is the best observed arm on top-10 recovery. It is a control, it was predeclared as a control, and it is reported as such. Promoting the best-observed control to primary after the fact would convert a diagnostic into a selected result.

11 Anti-leakage order and evaluation protocol

The execution order is fixed and one-directional:

1. All predictions are produced and written to disk.
2. Prediction artifacts are frozen under SHA256. The frozen prediction digest is
5bc46c27e260572cc33630ff59b3dd142cc37ca5ef66d13eb77f29b1a2ad0eda.
3. A *separate* entry point verifies every frozen digest, and only then imports the evaluator and reads gold and target annotations.

During step 1, prediction audit hooks deny reads of gold filenames, historical outputs, reports and notes, protected project source, and any evaluator import; permitted third-party packages are read-only. These hooks are a procedural discipline for ordinary Python execution; they are explicitly **not hostile native-code containment**. All local models carry fixed revisions and per-file hashes (Table 16), and model downloads require an explicit **-download** flag. The run uses the pre-existing interpreter’s third-party installation with **-B** and no runtime project imports from **rest_of_work**.

11.1 Metrics and denominators

The **strict direct denominator is 63**: all target occurrences across the 14 Type-A questions. Of these, 49 are directly mappable (48 **EXACT** + 1 **ALIAS**), 1 is **COMPOSITE**, 11 are **PARTIAL** and 2 are **ABSENT**. All 18 questions in the set have predictions; the 4 non-Type-A questions are not scored for direct identity.

Table 3: Denominators and target-occurrence status counts.

Quantity	Value
All questions (predictions produced)	18
Type-A questions (scored)	14
Target occurrences (strict denominator)	63
Directly mappable occurrences (EXACT + ALIAS)	49
EXACT	48
ALIAS	1
PARTIAL	11
ABSENT	2
COMPOSITE	1

Definitions, applied without exception:

$$R@k = \frac{|\{\text{directly mapped target occurrences at rank } \leq k\}|}{63}, \quad (12)$$

$$R^{\text{cond}}@k = \frac{|\{\text{directly mapped target occurrences at rank } \leq k\}|}{49}, \quad (13)$$

$$\text{MRR} = \frac{1}{14} \sum_{j=1}^{14} \frac{1}{r_j}, \quad \frac{1}{r_j} := 0 \text{ if question } j \text{ has no direct hit}, \quad (14)$$

where r_j is the rank of the *first* directly mapped identity for Type-A question j . The MRR denominator is 14 — all Type-A questions, including Q5 and Q11 which contain *no* directly mappable target and therefore contribute structural zeros.

Direct PARTIAL, ABSENT and COMPOSITE occurrences receive no identity credit under any arm. They remain in the denominator of 63 because removing them would flatter the system by discarding exactly the cases it cannot represent. Composite completion is tracked separately and is zero for every arm reported here: the single COMPOSITE target, Q11’s CE→MC→NN→PF pipeline, is never assembled.

Target-recovery counts are occurrence counts. MACE appears as a target occurrence in Q1, Q12 and Q13; each occurrence is a separate scored item with its own rank, because the same entity can be correct-and-well-ranked for one question and correct-but-buried for another. This convention applies to the recovery, recall, failure-class and reranking-movement counts. It does *not* apply to the index statistics of Table 1, which count distinct entities, evidence items, links and arcs, nor to efficiency counters, which count units of logical work.

Candidate recall uses mapped EXACT/ALIAS occurrences, reported alongside raw-denominator figures, at depths 50/100/200/500/1000, plus union recall and candidate set size. For reranked arms, pool membership is measured **before** reranking.

Rank statistics are conditional on *observed* direct ranks, with the missing count stated explicitly. A fixed missing-rank penalty is also saved; it is the same constant for every arm and is recoverable from the reported conditional mean, observed count and penalized mean (its value is 5799, and the identity $\bar{r}_{\text{pen}} \cdot 49 = \sum_{\text{obs}} r + n_{\text{miss}} \cdot 5799$ holds for all reported arms).

Positive controls and empty-rank controls are isolated under **evaluation_only**. They are **gold-informed evaluation instruments and never real predictions**; no reported system metric derives from them.

12 Results

12.1 Candidate recall: hypothesis H1

Candidate set size ranges are 349–500 for A1/TypedRaw/ParsedOnly/A6/A7, 349–1090 for A2, 349–1324 for A3 and WithoutGraphRRF, 349–1503 for A4/A5, 349–1362 for WithoutEvidenceRRF, and exactly 500, 100, 200 for A0, A6_100, A6_200.

On the recall question the observed development evidence is clear. A5 admits **37/49 at depth 100, 44/49 at depth 500, and 46/49 in the full union**, against **3/49 at depth 100** for the previous V2 high-recall arms and 2/49 for the previous V2 A0. The progression A1 → A2 → A3 → A4 shows that most of the gain comes from channel addition rather than from fusion: depth-100 hits move 10 → 20 → 31 → 35 under pure round-robin union. RRF then lifts depth-50 hits from 17 (A4) to 29 (A5) on the identical union, which is the one place where ordering alone is demonstrably the operative mechanism.

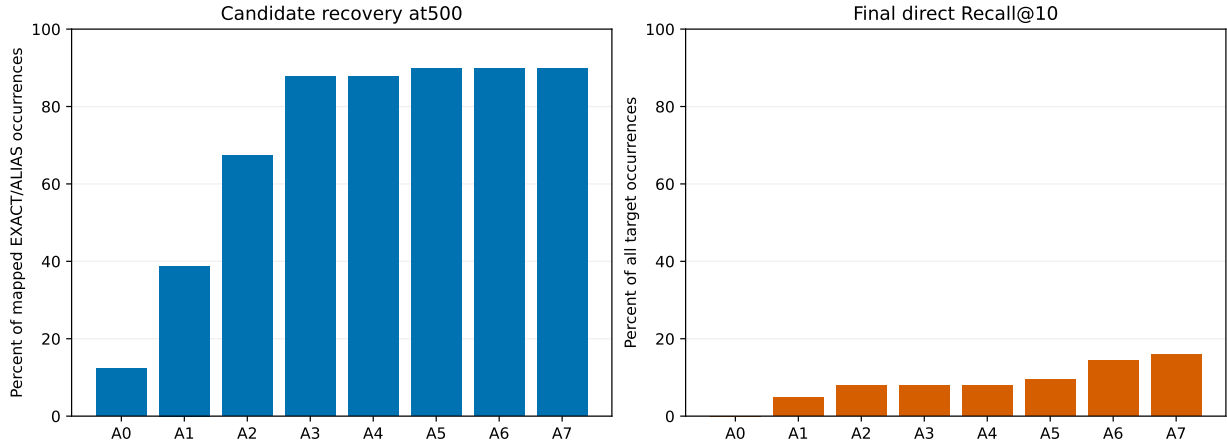


Figure 1: Two panels for the main stage sequence A0–A7. *Left*: candidate recovery at depth 500, as a percentage of the 49 mapped EXACT/ALIAS occurrences. *Right*: final direct $R@10$, as a percentage of all 63 target occurrences. The figure shows these two quantities only — no rank distributions and no additional cut-offs — and the controls are not plotted. All values are development diagnostics from the frozen artifacts.

Table 4: Candidate pool recall. Hits are directly mapped target occurrences out of 49; R^{cond} is the /49 fraction and R^{raw} the /63 fraction, both at depth 500. “Union” is the full candidate pool regardless of depth. For the reranked arms the candidate pool is the *actual reranking prefix*: A6_100 and A6_200 cap their pools at 100 and 200 and are saturated at greater depths, and A6/A7 cap at 500 (mean size 489.2 because some questions yield fewer than 500 eligible candidates). **No RRF tail is included in any candidate figure in this table**; the final rankings of those arms do append the unchanged tail, which is why their observed-rank statistics in Table 7 cover 46 occurrences while their candidate pools contain 37–44.

Arm	Hits at pool depth (/49)					Depth 500		Union	
	50	100	200	500	1000	R^{cond}	R^{raw}	hits	avg. size
A0	1	2	3	6	6	0.122	0.095	6	500.0
A1	5	10	16	19	19	0.388	0.302	19	489.2
TypedRaw	5	10	16	19	19	0.388	0.302	19	489.2
ParsedOnly	4	6	12	19	19	0.388	0.302	19	489.2
A2	11	20	24	33	43	0.673	0.524	43	913.6
A3	16	31	35	43	44	0.878	0.683	45	1108.9
A4	17	35	42	43	46	0.878	0.683	46	1295.6
A5	29	37	43	44	46	0.898	0.698	46	1295.6
WithoutEvidenceRRF	17	29	40	45	46	0.918	0.714	46	1149.5
WithoutGraphRRF	18	26	35	44	44	0.898	0.698	45	1108.9
A6_100	29	37	37	37	37	0.755	0.587	37	100.0
A6_200	29	37	43	43	43	0.878	0.683	43	200.0
A6 (=A6_500)	29	37	43	44	44	0.898	0.698	44	489.2
A7	29	37	43	44	44	0.898	0.698	44	489.2

Channel ablations

Removing the evidence list from fusion drops A5’s depth-50 hits from 29 to 17 and depth-100 from 37 to 29. Removing the graph list drops the same figures to 18 and 26. Both lists matter at shallow depth and **neither is categorically more important**: the evidence list is worth more at depth 50 (12 versus 11 occurrences) and the graph list is worth more at depth 100 (11 versus 8). Note also that `WithoutEvidenceRRF` has the *highest* depth-500 recall of any arm (45/49) while sharing A5’s union recall of 46/49, so that difference is a reordering effect inside the 500-prefix, not a loss of coverage.

Table 5: Per-list diagnostics on the A4/A5 union. “Unique” = mapped occurrences recovered by that list and no other.

Ranked list	Mapped occurrences recovered	R^{cond} (/49)	Unique
Dense (original question)	19	0.388	0
Dense (structured question)	21	0.429	0
BM25	41	0.837	0
Name	0	0.000	0
Evidence	44	0.898	0
Graph	37	0.755	1

Table 5 is the most informative single diagnostic in the study. The name list recovers **nothing**: these questions describe capabilities, not names, so exact-identity matching has no opportunity. BM25 over the concatenated lexical document recovers 41/49 — more than either dense view — so on this corpus the dense encoder, used without an instruction prefix and over bounded node renderings, is weaker than term matching. The evidence list is the single strongest list at 44/49. Only the graph list contributes anything *uniquely* (one occurrence), so in terms of pure membership the lists are largely redundant *collectively*: nearly every occurrence found by the evidence list is also found by some other list. That redundancy is about the union, not about ordering, and it is not the same as being useless at a given depth: adding the evidence list to the round-robin union (A2 $\rightarrow$ A3, before graph was added) raised depth-100 hits from 20 to 31. Membership redundancy with rank-level contribution is exactly the regime Equation (10) is designed for.

12.2 Final ranking metrics: hypothesis H2

As fractions of 63, the declared primary A7 achieves $R@1 = 1.59\%$, $R@5 = 7.94\%$, $R@10 = 15.87\%$, $R@25 = 22.22\%$, $R@50 = 34.92\%$, $R@100 = 52.38\%$. On the conditional /49 denominator the corresponding figures are 2.04%, 10.20%, 20.41%, 28.57%, 44.90%, 67.35%.

Four observations:

1. **Top-10 recovery is higher than in the prior iterations.** A7 recovers 10/63 (15.87%), against 3/63 for the previous V2’s dense baseline arm, 2/63 for its relevance arm V2-C, 1/63 for V2-B, and 0/63 for A0 in the matched re-run. This is the main evidence bearing on H2, and it compares whole systems, not the cross-encoder in isolation.
2. **The best observed arm is a control.** A6_100 reaches 12/63 at top-10 and 18/63 at top-25, above both A6 (9/63, 14/63) and A7 (10/63, 14/63), and it has the best conditional mean rank (77.70) of the reranked arms. It does *not* sweep every statistic: A6_200 has a marginally better all-observed median (39.5 versus 40.0). The declared primary remains A7.
3. **No single ordering of the arms holds across metrics.** A4’s round-robin union has the highest MRR of any arm at 0.2013; RRF reordering lowers it to 0.1554 (A5), cross-encoder

Table 6: Final direct identity recovery. Hits are target occurrences out of 63; $R@k$ fractions use the strict denominator 63. MRR denominator is 14.

Arm	Hits at rank $\leq k$ (/63)						MRR
	1	5	10	25	50	100	
A0	0	0	0	0	1	2	0.0044
A1	2	3	3	3	5	10	0.1666
TypedRaw	2	3	3	3	5	10	0.1667
ParsedOnly	0	2	3	3	4	6	0.0843
A2	2	3	5	6	11	20	0.1872
A3	2	4	5	10	16	31	0.1979
A4	2	4	5	10	17	35	0.2013
A5	0	4	6	11	29	37	0.1554
WithoutEvidenceRRF	0	3	3	8	17	29	0.1278
WithoutGraphRRF	0	4	4	8	18	26	0.1374
A6_100	1	6	12	18	25	37	0.1669
A6_200	1	5	9	16	26	34	0.1523
A6 (=A6_500)	1	5	9	14	22	33	0.1480
A7 (primary)	1	5	10	14	22	33	0.1516

reranking to 0.1480 (A6), and verification tiering recovers part of it at 0.1516 (A7), so MRR along the pipeline is non-monotonic. Meanwhile A5 raises top-50 hits from 17 to 29 and loses both of A4’s rank-1 hits ($R@1$: $2/63 \rightarrow 0/63$). Whether A5 or A4 is “better” depends entirely on the operating depth, and no arm here is better on every reported metric.

4. **Larger reranking pools were worse on top-10 in this run.** Top-10 hits fall 12 (pool 100), 9 (200), 9 (500), and the conditional median after reranking rises $27 \rightarrow 38 \rightarrow 50.5$. The straightforward reading is precision dilution: a bigger prefix gives the cross-encoder more opportunities to promote a plausible-looking non-target above a target. This is an observed pattern over three pool sizes in one run, not an established scaling law.

Table 7: Observed-rank statistics over the final rankings. Mean and median are conditional on observed direct ranks; the penalized mean assigns the fixed missing-rank penalty (5799) to each missing occurrence and divides by 49.

Arm	median	mean	observed	missing	penalized mean
A0	208.0	220.00	6	43	5115.86
A1	97.0	118.68	19	30	3596.43
TypedRaw	97.0	121.58	19	30	3597.55
ParsedOnly	161.0	169.37	19	30	3616.08
A2	128.0	257.95	43	6	936.45
A3	70.0	159.18	45	4	619.57
A4	60.5	97.80	46	3	446.86
A5	39.5	80.85	46	3	430.94
WithoutEvidenceRRF	86.0	109.57	46	3	457.90
WithoutGraphRRF	76.0	139.51	45	4	601.51
A6_100	40.0	77.70	46	3	427.98
A6_200	39.5	82.22	46	3	432.22
A6 (=A6_500)	51.5	106.54	46	3	455.06
A7	51.5	106.24	46	3	454.78

Table 7 shows why conditional statistics must be paired with miss counts. A0’s conditional mean rank of 220 looks respectable until one reads that it is computed over 6 observed occurrences with 43 missing; its penalized mean of 5115.86 is the honest summary. Conversely A2’s conditional

mean (257.95) is *worse* than A1’s (118.68) while A2 observes 43 occurrences to A1’s 19, and those extra 24 occurrences sit deep in the list. The penalized means — 936.45 for A2 against 3596.43 for A1 — order the two the way the recall figures do. Note that the reranked arms observe 46 occurrences here while their candidate pools in Table 4 contain 37–44: the extra occurrences live in the appended, unreranked RRF tail.

Table 8: Previous V2 (archived) arms, same occurrence denominators. Candidate hits are out of 49. Rank statistics there were computed over the full corpus, so misses are zero by construction.

Arm	$R@10$ hits	$R@100$ hits	MRR	median direct rank	cand. @100	cand. @200
prev. A0	0	2	0.0047	3122	2	3
prev. A1	3	6	0.1645	1293	6	8
V2-B	1	3	0.0168	1192	3	6
V2-C	2	3	0.0566	1192	3	6

These historical comparisons are **descriptive**. The hit counts and MRRs share the denominators used here and are directly readable. The rank medians are not comparable to Table 7: the archived arms ranked the whole corpus, so every target had a rank and the miss count was zero, whereas the new arms truncate candidates and report conditional means with explicit misses. Both are internally valid; they measure different things, and comparing them without the miss counts would be meaningless rather than merely imprecise. The primary contrasts in this report are the *matched new-run ablations* of Tables 4 and 6, computed under identical denominators and identical candidate caps.

12.3 Conditional reranking analysis

Conditional reranking statistics are computed only over target occurrences actually admitted to the reranker’s prefix; occurrences in the untouched tail cannot be moved and would dilute the measurement.

Table 9: Conditional reranking effects, restricted to target occurrences admitted to each prefix. “New top-10” counts occurrences that were outside top-10 before reranking and inside after.

Arm	pool	seen	up	down	median before	median after	top-10 after	new top-10
A6_100	100	37	17	20	36.0	27.0	12	7
A6_200	200	43	19	24	39.0	38.0	9	5
A6	500	44	18	26	39.0	50.5	9	5
A7	500	44	18	26	39.0	50.5	10	6

At pool size 500, **of the 44 target occurrences seen by the reranker, 18 moved up and 26 moved down**, and the conditional median moved from 39 to 50.5. Conditional top-10 recall is 0.2273 for A7 and 0.2045 for A6, against 0.3243 for A6_100. So at pool 500 the cross-encoder lowered more target occurrences than it raised while still placing more of them in the top 10 than fusion alone did: it concentrates a handful of occurrences near the top and pushes the majority down. Both facts are measured and must be reported together.

The verification tier (A7 versus A6) changes one occurrence’s top-10 status and one question’s first direct rank in the frozen artifacts: in Q12, **DeePMD** moves from rank 20 to rank 7 and **GAP** from rank 5 to rank 4, so Q12’s first direct rank improves from 5 to 4 and A7 gains one top-10 occurrence (6 new top-10 against 5 for A6). Every other per-question *first* direct rank is identical between A6 and A7. That is not the same as saying only one occurrence moved: tier reordering is applied to the whole top 20, so other candidates — including non-target entities and target occurrences that stayed outside the top 10 — may have changed position inside that window without changing any reported first-rank statistic. The measured *gain* attributable to the stage

on this data is one top-10 occurrence, with MRR moving $0.1480 \rightarrow 0.1516$. The robustness of that gain is unmeasured: this is a single seeded run, no variance estimate exists, and no hypothesis test was performed, so nothing here either establishes or excludes that a rerun or a different question set would reproduce it.

12.4 Selected strongest movements

The following tables reproduce exact before/after ranks from the frozen handoff. They are presented as **movements**, not explanations: the artifacts record ranks, not causes, and no causal attribution to evidence similarity, card content or token truncation is inferred here. Rank movement also does not adjudicate scientific relevance: an entity promoted above a target is not thereby shown to be a worse answer, and a non-gold entity ranked highly is not thereby shown to be scientifically wrong.

Table 10: Strongest rank improvements and degradations under cross-encoder reranking at pool 500 (identical for A6 and A7). Ranks are exact handoff values.

Improvements				Degradations			
Q	Target	before	after	Q	Target	before	after
Q9	NequIP	256	60	Q12	MACE	42	302
Q13	CGCNN	118	17	Q12	PET	61	317
Q13	MACE	136	50	Q7	symbolic regression	48	276
Q13	CHGNet	113	35	Q1	ACE	159	370
Q10	SchNet	156	87	Q4	Hessian training	44	229

Table 11: Strongest rank improvements and degradations under cross-encoder reranking at pool 100 (A6_100). Ranks are exact handoff values.

Improvements				Degradations			
Q	Target	before	after	Q	Target	before	after
Q12	CHGNet	94	20	Q12	MACE	42	86
Q1	CHGNet	72	15	Q12	eSEN	12	53
Q12	DeePMD	62	8	Q14	WBM	29	68
Q12	GAP	50	4	Q1	MACE	39	74
Q12	EquiformerV2	57	19	Q12	PET	61	89

Two patterns are worth recording. **MACE** is degraded in both Q1 and Q12 at every pool size (Q12: $42 \rightarrow 86$ at pool 100, $42 \rightarrow 157$ at pool 200, $42 \rightarrow 302$ at pool 500), while **MACE** in Q13 is *improved* at pools 200 and 500 ($136 \rightarrow 40$ and $136 \rightarrow 50$). The same entity moves in opposite directions for different questions, which is consistent with question-conditional scoring rather than a per-entity bias. Separately, degradation magnitude grows with pool size for the same occurrence — Q12 PET at $61 \rightarrow 89 \rightarrow 161 \rightarrow 317$ for pools 100/200/500 — which is the occurrence-level counterpart of the pool-size pattern in Table 9.

12.5 Per-question summary

Q5 and Q11 have **no directly mappable target** and are structurally unanswerable under the direct identity metric: Q5’s targets are **Bead-mapping** (ABSENT) and **GNN+GPP** (PARTIAL); Q11’s are the **COMPOSITE** pipeline and **hybrid frameworks** (PARTIAL). They contribute two guaranteed zeros to every arm’s MRR, capping attainable MRR at $12/14 = 0.857$ even with perfect rank-1 performance elsewhere.

Table 12: Per-question first directly mapped rank. “T” = target occurrences; “D” = directly mappable (EXACT/ALIAS) occurrences. “-” = no direct hit anywhere in the list.

Q	T	D	A0	A1	A5	A6	A7
Q1	7	7	—	193	23	3	3
Q2	3	2	369	85	6	10	10
Q3	2	1	57	1	2	7	7
Q4	3	2	—	37	33	69	69
Q5	2	0	—	—	—	—	—
Q6	3	1	199	4	2	42	42
Q7	2	1	—	—	48	276	276
Q8	2	1	—	—	7	6	6
Q9	3	1	—	—	256	60	60
Q10	2	1	—	—	156	87	87
Q11	2	0	—	—	—	—	—
Q12	13	13	31	1	2	5	4
Q13	8	8	—	—	89	17	17
Q14	11	11	217	26	4	1	1
Total	63	49					

At the first-hit level, reranking helps and hurts in comparable measure. Relative to A5, A6/A7 improve Q1 (23 $\rightarrow$ 3), Q8 (7 $\rightarrow$ 6), Q9 (256 $\rightarrow$ 60), Q10 (156 $\rightarrow$ 87), Q13 (89 $\rightarrow$ 17) and Q14 (4 $\rightarrow$ 1), and degrade Q2 (6 $\rightarrow$ 10), Q3 (2 $\rightarrow$ 7), Q4 (33 $\rightarrow$ 69), Q6 (2 $\rightarrow$ 42), Q7 (48 $\rightarrow$ 276) and Q12 (2 $\rightarrow$ 4, or 2 $\rightarrow$ 4 for A7 after tiering). The Q6 case is the sharpest: LiFlow sits at rank 2 after fusion and falls to 42 after reranking.

12.6 Failure analysis

Every target occurrence receives one documented **primary** failure class at top-10, plus optional secondary observations. The primary classes partition all 63 occurrences exactly.

Table 13: Primary failure classification at top-10 over all 63 target occurrences, and secondary observations (non-exclusive). Codes are used in Appendix B.

Code	Class	Count	Meaning
CE	IN_POOL_CROSS_ENCODER_FAILED	34	Occurrence was inside the 500-deep reranking prefix and is not in the final top 10.
R10	RECOVERED_AT_10	10	Recovered at top-10 by the primary system.
GT	GROUND_TRUTH_MAPPING_UNCERTAIN	14	Non-direct annotation status (PARTIAL/ABSENT/COMPOSITE); no identity credit possible.
RRF	IN_POOL_RRF_FAILED	2	In the union but ranked outside the 500-deep prefix by RRF, so never reranked.
TF	WRONG_TYPE_FILTER	3	Excluded by the hard answer-family filter.
	Total	63	
NN	NO_NAME_MATCH (secondary)	46	No lexical opportunity: the question contains no surface form of the target.
NE	NO_RELEVANT_EVIDENCE (secondary)	8	No indexed candidate-specific evidence existed.

The largest class, IN_POOL_CROSS_ENCODER_FAILED at 34 of 63 occurrences, must be read exactly as defined: these occurrences were *inside the prefix and are not in the final top 10*. The class does **not** mean 34 occurrences were actively demoted or lost by the cross-encoder; most of

them were far from the top 10 before reranking as well. The measured movement attributable to the reranker is the conditional analysis of Table 9: of the 44 admitted occurrences, 18 moved up and 26 moved down. Nor does the small size of the fusion (2) and type-filter (3) classes license the conclusion that admission has ceased to be a constraint — five mapped occurrences never reached the reranker at all, including three outside the union entirely, and 14 occurrences cannot be credited by any system. What the distribution supports is narrower and still useful: at these depths, with this pool, most remaining error is located after admission, in the ordering stages.

Concrete named traces, all from the frozen handoff:

- **Q13** MEGNet and **Q13** EquiformerV2 are the two IN_POOL_RRF_FAILED cases, at ranks 733 and 551 respectively in A5, A6 and A7 alike. They are in the union but beyond the 500-deep prefix, so their ranks are identical across all three arms by construction: the tail is never touched.
- **Q14** JARVIS-DFT, **Q14** AFLOW and **Q14** Alexandria are the three WRONG_TYPE_FILTER cases, with null ranks under every arm. Q14 requests “datasets and benchmarks” (`{dataset,metric}`), and the graph nodes bearing these names are typed outside those families, so Equation (2) removes them before retrieval. Three of Q14’s eleven occurrences are lost this way.
- **Q12** MACE is the largest single downward movement: rank 42 in A5, 302 in A6 and A7. **Q12** PET follows the same trajectory, 61 $\rightarrow$ 317; **Q7** symbolic regression 48 $\rightarrow$ 276; **Q1** ACE 159 $\rightarrow$ 370.
- **Q1** GAP is a clean recovery: null in A0 and A1, rank 31 in A5, rank 3 in A6/A7. **Q14** Matbench Discovery moves 447 (A0) $\rightarrow$ 97 (A1) $\rightarrow$ 4 (A5) $\rightarrow$ 1 (A6/A7), and **Q14** Matbench 157 $\rightarrow$ 39 $\rightarrow$ 3. **Q8** ConvLSTM is null in A0 and A1, rank 7 in A5 and 6 in A6/A7 — an occurrence the dense-only arms could not reach at all.
- **Q12** DeepMD is the occurrence whose top-10 status is attributable to the verification stage: rank 62 in A5, 20 in A6, 7 in A7.
- **Q2** HamGNN (PARTIAL), **Q3** xDeepH (PARTIAL), **Q4** MACE-F (ABSENT) and **Q11** CE $\rightarrow$ MC $\rightarrow$ NN $\rightarrow$ PF (COMPOSITE) illustrate the 14 GROUND_TRUTH_MAPPING_UNCERTAIN occurrences: null under every arm by annotation status rather than by retrieval behaviour.

Two terminological commitments. NO_RELEVANT_EVIDENCE means *no indexed candidate-specific evidence existed* — an index coverage statement, not an adjudication that the entity is scientifically irrelevant to the question. NO_NAME_MATCH means *the absence of a lexical opportunity* — the question never names the target — not that name matching was attempted and failed.

12.7 Missing evidence versus contradiction

Of 800 requirement components, only 376 had an eligible premise; the remaining 424 are NOT_SUPPORTED_BY_AVAILABLE_EVIDENCE purely because the candidate-specific index, with article-level and structural-path evidence excluded, contained nothing to score. Adding the 270 neutral NLI outcomes gives the reported total of 694.

The status counts aggregate positive and exclusion components, whose polarities are opposite under Equation (11). The 71 SUPPORTED statuses therefore comprise positive components with $P(\text{entail}) \geq 0.7$ and $P(\text{contra}) < 0.3$ together with exclusion components with $P(\text{contra}) \geq 0.7$ and $P(\text{entail}) < 0.3$; symmetrically for the 35 CONTRADICTED statuses. Reading the aggregate as a single entailment-probability condition would be wrong unless exclusion components happen to be absent, which the handoff does not report.

Table 14: Requirement verification outcomes over the A7 top-20 candidates across all 18 questions.

Quantity	Value
Verification candidates (18×20)	360
Requirement components evaluated	800
NLI inference pairs actually run	376
of which SUPPORTED	71
of which CONTRADICTED	35
neutral / no clear result	270
Components with no eligible premise (inference skipped)	424
NOT_SUPPORTED_BY_AVAILABLE_EVIDENCE (total)	694

With that caveat, one distinction is substantive. **53 % of all components (424/800) are unsupported for lack of a premise, not because anything contradicts them.** Only 35 components — 4.4 % of the total, 9.3 % of actually inferred pairs — show high proxy contradiction. A design that treated missing evidence as contradiction would demote the majority of top-20 candidates on the basis of index sparsity rather than evidence, and given that 2508 of 2978 entities have no candidate-specific evidence at all, that error would systematically favour the well-documented entities.

Limitations of the NLI proxy:

- The probabilities are **uncalibrated**. The threshold $\tau = 0.7$ is a fixed design constant, not a tuned or validated operating point, and no reliability diagram supports it.
- **nli-deberta-v3-small** is a small general-domain model applied to materials-science claims far outside its training distribution. It can misread quantitative conditions (“millions of atoms”, “ $> 10^3$ atoms”, “ $\sim 50,000$ compounds”) and it can misread negation and scope.
- A premise is a single max-cosine evidence item, so verification is single-premise: a requirement supported jointly by two pieces of evidence will not be recognised.
- **This is not scientific entailment.** A **SUPPORTED** status means a small proxy model assigned a high probability to one retrieved sentence under the polarity of Equation (11); it is not an adjudication that the method satisfies the requirement.
- The measured gain of the whole stage on this data is one top-10 occurrence, with other movements inside the top 20 not separately reported. Its robustness is unmeasured.

12.8 Efficiency

All counters describe **logical work**, not production latency, and all timings are descriptive CPU wall times on a single machine with 4 threads, batch size 32, seed 42.

Three caveats are material. First, the BM25 counter embedded in the predictor is a *token-count estimate*; the 1,637,900 figure is the document-term loop count implied by the frozen query intent and the frozen tokenizer (entities $\times$ unique query terms, summed over questions), and it is the figure that should be cited. Second, the prefix timing estimates for pools 100 and 200 are **amortized reuses of the 500-run pair scores**: those controls never executed their own inference, so 170.12s and 325.14s are reconstructions of what the measured 500-pool pairs imply, not observations. Third, cross-encoder pairs equal actual inference pairs (8,849), confirming that no caching shortcut reduced the measured reranking workload in the 500 run.

On these two axes together, **A6_100** was both the best observed arm on top-10 recovery and the cheapest reranking configuration, by roughly a factor of four in prefix time.

Table 15: Efficiency counters and descriptive timings. Counters are units of logical work, not target occurrences. Timings are single-run CPU wall times.

Quantity	Value
Prediction phase wall time (s)	865.24
Query wall time (s)	745.26
Dense dot products	295,920
Evidence generation comparisons	153,864
BM25 document-term lookups (exact loop count)	1,637,900
Cross-encoder pairs	8,849
Cross-encoder actual inference pairs	8,849
Requirement evidence comparisons	170,960
Verification candidates	360
Verification components	800
NLI pairs	376
Prefix timing estimate, pool 100 (s)	170.12
Prefix timing estimate, pool 200 (s)	325.14
Prefix timing estimate, pool 500 (s)	675.01

13 Limitations

1. **Development-only, design-exposed.** The questions and gold annotations were seen during V1 development, and that exposure motivated this work. Execution-time separation does not make this a held-out result. Nothing here supports a publication claim or an inferential comparison.
2. **Scale.** 14 scored questions, 63 target occurrences, 49 directly mappable. One occurrence is 1.6 percentage points of $R@k$. The A7-versus-A6_100 top-10 gap is two occurrences.
3. **Annotation ceiling.** 14 of 63 occurrences (22 %) cannot receive identity credit under any system, and two of 14 questions have no creditable target at all. Maximum attainable $R@k$ is $49/63 = 77.8\%$; maximum attainable MRR is $12/14 = 0.857$.
4. **Index sparsity (schema coverage).** 2508 of 2978 entities have no candidate-specific evidence. The evidence channel, the graph channel’s evidence seeding, the cross-encoder card and the verification stage are all structurally weaker for those entities. This is a coverage property of the graph. Those entities remain reachable by the dense channels and by BM25 over names and aliases.
5. **Schema inference is structural.** Role arcs come from unique typed member signatures; ambiguous **extends** arcs and signature mismatches are dropped rather than resolved, so some legitimate paths are unavailable, and citation paths provide structural context and are never treated as entailment.
6. **Year-granular time filtering.** “By Feb 2026” is enforced at year granularity; month precision is unrecoverable from year-only evidence. **first_seen_year** records publication visibility in this graph, not an actual discovery date, and the undated-alias suppression rule only applies at cutoffs earlier than a node’s last observation.
7. **No calibration and no tuning.** Cross-encoder logits are not probabilities; NLI probabilities are uncalibrated. Path decay (0.8), RRF k (60), evidence/graph depths (500/200), family caps, the 512-token budget and the NLI threshold (0.7) are fixed design constants; no hyperparameter search was performed, so none is claimed optimal. The **soft_type_prior** key in **config.json** is legacy and unused.

8. **Controls beyond their prefix.** For A6_100 and A6_200, final ranks beyond the pool size come from the unchanged RRF tail, not from reranked entities, and their candidate figures in Table 4 are capped at the pool. Comparing $R@100$ across pool sizes compares different objects below the prefix boundary.
9. **Single run, no variance.** Every number is one seeded execution. No variance estimate and no hypothesis test exists, so differences of one or two occurrences are reported as observations only.
10. **Ranks are not scientific adjudication.** The metrics score agreement with a small annotation set. A high-ranked non-gold entity is not shown to be a wrong answer, and a low-ranked gold entity is not shown to be irrelevant.
11. **Historical comparisons are descriptive.** Prior V1 and V2 figures were computed under different candidate truncation and grouping conventions; their rank statistics are not commensurate with the conditional statistics here.

14 Future work

Two questions are open, and neither can be settled by rerunning arms that already exist in this study. A6_100 is such an arm: it is reported above as a predeclared control, so a “cascade that reranks only the top 100” is a configuration already measured here, not a new experiment, and quoting its 12/63 as a prediction would be reporting a known result.

1. **Pool choice needs independent data.** The pool-size pattern (top-10 hits 12/9/9; conditional median 27/38/50.5) comes from the same 14 design-exposed questions used to build the system. The appropriate next step is to assemble an *independent* question and target set — new questions, annotated without reference to this system’s outputs — and to register the pool-size comparison on that set before running it. Until then the pool choice is a development observation.
2. **Evidence packing is a hypothesis, not a finding.** The large downward movements (Q12 MACE 42 $\rightarrow$ 302, Q12 PET 61 $\rightarrow$ 317) have exported card contents: included blocks, omitted blocks and token counts are already in the frozen artifacts. The honest sequence is to *inspect those existing traces first*, descriptively, and only then preregister a comparison of alternative evidence-packing policies — different family caps, different block order, best-fit instead of prefix packing — on new data. No such comparison is predeclared here, and no tuning of packing on the present benchmark should be presented as a held-out result afterwards.

Nothing in the present data justifies extending the verification stage; its measured gain is one occurrence and should be held fixed while the questions above are addressed. The three `WRONG_TYPE_FILTER` occurrences are a separate, concrete item: widening Q14-style coordinated answer families would be a design change whose cost in admitted noise is currently unmeasured.

15 Reproducibility

The following links identify the pinned local artifacts only; no external source was consulted for this report: <https://huggingface.co/BAAI/bge-small-en-v1.5>, <https://huggingface.co/cross-encoder/ms-marco-MiniLM-L6-v2>, <https://huggingface.co/cross-encoder/nli-deberta-v3-small>.

Table 16: Pinned model versions. All models are loaded locally from hashed caches; downloads require an explicit `-download` flag. Revisions wrap across lines and concatenate to the exact literal value.

Role	Model	Revision	Status
Dense	BAAI/bge-small-en-v1.5	5c38ec7c405ec4b44b94 cc5a9bb96e735b38267a	available
Relevance	cross-encoder/ ms-marco-MiniLM-L6-v2	233902d25c440f23af6f 7d6e94d2946bac0bee0a	available
NLI (verification only)	cross-encoder/ nli-deberta-v3-small	fa2804872c3b4bd748f3 8c0185cc85775361e735	available

Environment: Python 3.11.2 (MSC v.1934, 64-bit), CPU device, seed 42, 4 threads, batch size 32; `numpy` 2.3.5, `torch` 2.4.1, `transformers` 4.57.6, `sentence-transformers` 5.1.0, `huggingface-hub` 0.36.2. No model load failures were recorded. All per-file SHA256 digests for the three model caches are recorded in `outputs/audit/model_version_manifest.json`.

15.1 Two distinct determinism checks

These are different checks with different scopes and are not interchangeable:

1. **Replay check.** The replay reuses *cached model outputs* and verifies that the full prediction byte stream reproduces identically against the frozen digest `5bc46c27e260572cc33630ff59b3dd142cc37ca5ef66d13eb77f29b1a2ad0eda`. This establishes byte-identical replay of the deterministic pipeline logic — grouping, filtering, union construction, RRF ordering, tie-breaking, card packing, tier assignment, serialization — *given the cached model outputs*. It does not re-execute the neural models and therefore says nothing about reproducibility of an uncached full-model run.
2. **Fixed-input model determinism tests.** Separate synthetic tests feed fixed inputs to the pinned *dense encoder* and the pinned *relevance cross-encoder* and check that the inference output is bit-stable, with deterministic algorithms enabled, when those models are available. The **NLI verification model is not covered** by this check; the NLI model is covered only by the status-semantics and policy tests, which assert that it is never used as a relevance scorer.

Together they establish deterministic pipeline logic given cached outputs, and bit-stable inference for two of the three models given identical inputs. Neither establishes — nor is claimed to establish — stability across hardware, library versions, uncached reruns, or re-trained checkpoints.

15.2 Scope of the change

The active implementation is `clean_tshc_v2`. The previous V2 source, config, report and results are archived under `archive/flat_20261001`. The experiment does **not** route through the hierarchy component and does not place it on any critical path.

Regarding V1: at the start of this work, V1 already contained two pre-existing uncommitted tracked report changes and four untracked report files, and the user explicitly approved **preserving that starting state**. The commitment made and kept is therefore that **this work introduces zero additional changes to V1** — not that `git diff` in V1 is empty. V1 hashes and diff should be compared against that approved starting state, not against `HEAD`. No GitHub operations of any kind were performed.

15.3 Verification and test tallies

Digest verification and the automated test suite are executed by Codex; the audited tallies are reproduced here. The test run observed at the time of writing was 42 tests passing in 16.35 s.

Final verification by Codex

The complete V2 suite passed: **42 tests passed in 16.35 seconds**, including existing regression tests and the new entity-pipeline and frozen-artifact checks. Prediction/source/input hashes verify, and the full cached replay has the same prediction SHA256. The positive and empty-ranking evaluation controls pass. V1 has **zero new modifications**: all 1,056 file hashes, the binary git diff, and V1 git-status entries match the user-approved starting state. Existing V1 report changes were preserved. No GitHub changes were made. The machine-readable verification, changed/added-file inventory, and final artifact hashes are under `outputs/audit/`. These checks do not turn development diagnostics into held-out evidence.

16 AI contribution disclosure

- **Codex** implemented the pipeline, integrated all components, executed the experiment, produced every measurement, supplied the factual corrections incorporated into this revision, and performs the numerical verification, digest checks, test runs and compilation of this report.
- **Claude Opus 5** supplied a methodology consultation — on metric denominators, failure classification, leakage ordering and the separation of the two determinism checks — authored this report from the frozen measured artifacts (`notes/entity_redesign/method_specification.md`, `notes/entity_redesign/report_handoff.json`, `config.json`, `outputs/audit/model_version_manifest.json` and the pipeline source), and revised it against Codex’s corrections. Claude did not run the experiment, did not produce any measurement, and did not select which arm is primary.

This report is not human expert adjudication and not a held-out publication result. It is a development diagnostic produced by AI agents from frozen artifacts, over a question set that was visible during earlier design iterations, at a scale (14 questions, 63 target occurrences) where single-occurrence differences dominate the reported margins. The declared primary system is A7, as predeclared, and is reported as such even though a predeclared control performs better on the headline metric.

A Full question texts

ID	Question (verbatim)
Q1	Which methods (by Feb 2026) are best suited for predicting interatomic potentials when balancing accuracy close to DFT with computational efficiency for systems of millions of atoms?
Q2	Which methods (by Feb 2026) are best suited for accelerating electronic structure calculations when self-consistent field iterations become prohibitively expensive for large systems ($> 10^3$ atoms)?

ID	Question (verbatim)
Q3	Which methods (by Feb 2026) are best suited for learning DFT Hamiltonians when dealing with twisted van der Waals heterostructures containing thousands of magnetic atoms?
Q4	Which methods (by Feb 2026) are best suited for phonon property prediction when universal MLIPs show inaccuracies despite good energy/force performance?
Q5	Which methods (by Feb 2026) are best suited for coarse-graining molecular dynamics when crystalline symmetries and long-range elastic interactions must be preserved?
Q6	Which methods (by Feb 2026) are best suited for predicting ionic conductivity when full ab initio MD is too expensive but accuracy must be maintained?
Q7	Which methods (by Feb 2026) are best suited for predicting phase transitions when screening $\sim 50,000$ inorganic compounds at high throughput?
Q8	Which methods (by Feb 2026) are best suited for predicting fracture patterns when generalizing from atomistic MD data to unseen bicrystalline structures?
Q9	Which methods (by Feb 2026) are best suited for ensuring physical consistency when transferring information between scales without violating thermodynamic principles?
Q10	Which methods (by Feb 2026) are best suited for representing atomic environments when the system exhibits significant compositional diversity, defects, and disorder?
Q11	Which methods (by Feb 2026) are best suited for integrating features from three or more distinct scales simultaneously within a unified framework?
Q12	What are the leading (by Feb 2026) ML-based interatomic potential methods for large-scale atomistic simulation?
Q13	What are the leading (by Feb 2026) GNN architectures for predicting properties of solid-state crystalline materials?
Q14	What datasets and benchmarks are commonly used (by Feb 2026) to evaluate MLIPs and property prediction models for materials discovery?

B Per-occurrence target ranks

Annotation statuses: E = EXACT, A = ALIAS, P = PARTIAL, X = ABSENT, C = COMPOSITE. Failure codes are those of Table 13: CE, R10, GT, RRF, TF. “–” = not retrieved by that arm.

Q	Target	St	A0	A1	A5	A6	A7	Fail
Q1	MACE	E	–	–	39	174	174	CE
Q1	M3GNet	E	–	193	35	51	51	CE
Q1	CHGNet	E	–	309	72	31	31	CE
Q1	GAP	E	–	–	31	3	3	R10
Q1	NequIP	E	–	354	24	56	56	CE
Q1	ACE	E	–	–	159	370	370	CE
Q1	EquiformerV2	E	–	–	23	54	54	CE
Q2	DeepH	E	–	–	40	15	15	CE
Q2	D4FT	E	369	85	6	10	10	R10
Q2	HamGNN	P	–	–	–	–	–	GT
Q3	xDeepH	P	–	–	–	–	–	GT
Q3	DeepH-E3	E	57	1	2	7	7	R10
Q4	Hessian training	A	–	37	44	229	229	CE
Q4	VGNN	E	–	–	33	69	69	CE
Q4	MACE-F	X	–	–	–	–	–	GT
Q5	Bead-mapping	X	–	–	–	–	–	GT

Q	Target	St	A0	A1	A5	A6	A7	Fail
Q5	GNN+GPP	P	–	–	–	–	–	GT
Q6	DeePMD+AL	P	–	–	–	–	–	GT
Q6	LiFlow	E	199	4	2	42	42	CE
Q6	MLIP descriptors	P	–	–	–	–	–	GT
Q7	GNN free energies	P	–	–	–	–	–	GT
Q7	symbolic regression	E	–	–	48	276	276	CE
Q8	ConvLSTM	E	–	–	7	6	6	R10
Q8	GNN stress fields	P	–	–	–	–	–	GT
Q9	E(3)-equivariant GNNs	P	–	–	–	–	–	GT
Q9	physics-informed	P	–	–	–	–	–	GT
Q9	NequIP	E	–	–	256	60	60	CE
Q10	GNNs + ACE descriptors	P	–	–	–	–	–	GT
Q10	SchNet	E	–	–	156	87	87	CE
Q11	CE→MC→NN→PF	C	–	–	–	–	–	GT
Q11	hybrid frameworks	P	–	–	–	–	–	GT
Q12	MACE	E	–	–	42	302	302	CE
Q12	NequIP	E	–	–	39	65	65	CE
Q12	CHGNet	E	–	252	94	38	38	CE
Q12	M3GNet	E	–	141	36	108	108	CE
Q12	SevenNet	E	–	–	40	50	50	CE
Q12	GAP	E	–	–	50	5	4	R10
Q12	DeePMD	E	–	–	62	20	7	R10
Q12	MatterSim	E	31	1	2	24	24	CE
Q12	Orb	E	–	–	26	52	52	CE
Q12	eSEN	E	–	97	12	99	99	CE
Q12	PET	E	–	–	61	317	317	CE
Q12	UMA	E	–	–	67	161	161	CE
Q12	EquiformerV2	E	–	–	57	37	37	CE
Q13	CGCNN	E	–	–	118	17	17	CE
Q13	SchNet	E	–	–	115	198	198	CE
Q13	MEGNet	E	–	–	733	733	733	RRF
Q13	M3GNet	E	–	–	95	142	142	CE
Q13	NequIP	E	–	–	89	121	121	CE
Q13	MACE	E	–	–	136	50	50	CE
Q13	CHGNet	E	–	–	113	35	35	CE
Q13	EquiformerV2	E	–	–	551	551	551	RRF
Q14	Matbench Discovery	E	447	97	4	1	1	R10
Q14	Matbench	E	–	157	39	3	3	R10
Q14	JARVIS-DFT	E	–	–	–	–	–	TF
Q14	Materials Project	E	–	75	15	49	49	CE
Q14	MPtrj	E	–	108	37	16	16	CE
Q14	OQMD	E	–	120	33	67	67	CE
Q14	AFLOW	E	–	–	–	–	–	TF
Q14	Alexandria	E	–	–	–	–	–	TF
Q14	GNoME	E	–	69	26	10	10	R10
Q14	WBM	E	217	26	29	86	86	CE
Q14	OMat24	E	–	129	21	4	4	R10

C Fixed configuration constants

Snapshot 2026; seed 42; 4 threads; batch size 32. Channel depth 500 per list; RRF $k = 60$; BM25 $k_1 = 1.5$, $b = 0.75$; evidence seed depth 500; graph seed depth 200, article degree cap 80, fanout cap 64, max 2 hops, path decay 0.8; rerank prefix 500 with controls at 100 and 200; cross-encoder token budget 512; card family caps claims 4, explicit mentions 3, tasks 2, problems 2, techniques 2, components 2, datasets 1, metrics 1, articles 1, graph evidence 2; verification $k = 20$; NLI threshold 0.7; verification policy `stable_top20_tiers_all_supported_then_unresolved_then_contradicted`; diffusion arm not run. The key `soft_type_prior = 0.8` is present in `config.json` but unused by this pipeline.